Proposed Solution

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	5 Marks

Project Proposal Report — HDI Predictor

The proposal report aims to transform how researchers, policymakers, and analysts evaluate Human Development Index (HDI) scores. By building a machine learning-based web application, this project provides an accessible, interactive tool for real-time HDI prediction using key socio-economic indicators.

Project Overview

Field	Details
Objective	Build a Flask-based web application that predicts HDI scores and classifies countries into Very High, High, Medium, or Low development tiers using a trained Linear Regression model.
Scope	End-to-end ML project covering data loading, EDA visualization, preprocessing, model training, evaluation, serialization, and deployment via Flask web interface.

Problem Statement

Field	Details
Description	No simple interactive ML tool exists for real-time HDI prediction. Existing solutions require manual computation or complex Python scripting, making them inaccessible to non-technical users.
Impact	Solving this gap enables policymakers, researchers, and students to instantly predict HDI scores and identify areas requiring development intervention, supporting better policy decisions.

Proposed Solution

Field	Details
Approach	Linear Regression model trained on 195-country HDI dataset. Model serialized with Pickle. Deployed via Flask with 3 HTML pages: Home, Prediction Form, Result Display.
Key Features	1. Country selection dropdown (Label Encoded) 2. Input fields for Life Expectancy, Mean Schooling, GNI per Capita, Internet Users 3. Real-time HDI score prediction 4. Automatic HDI category classification 5. Actual vs Predicted scatter plot visualization
Model Accuracy	R-squared Score: 93.74% on test dataset (20 countries, 10% split)

Resource Requirements

Resource Type	Description	Specification
Hardware — Computing	CPU specifications	Intel Core i3 or above, 4GB RAM minimum
Hardware — Storage	Disk space for data and model	10 GB free space minimum
Software — Frameworks	Web framework	Flask 3.x (Python WSGI)
Software — Libraries	ML & Data libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Software — IDE	Development environment	Jupyter Notebook, PyCharm
Data — Dataset	Training & test data source	HDI.csv — 195 rows, 82 columns, Kaggle/UNDP
Data — Model File	Serialized model	HDI.pkl — 732 bytes (Pickle format)